



REVIEW ARTICLE

Ultraviolet–C radiation for sustainable control of harmful algal blooms and aquatic weeds

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Received: 28 April 2026; Accepted: 15 June 2026; Available online: Version 1.0: 14 September 2026

Cite this article: Venuaravind C, Thirukumaran K, Karthikeyan R, Suganthi M, Vellaikumar S, Umapathi M, Mahachandramuki E, Parthiban R J. Ultraviolet C radiation for sustainable control of harmful algal blooms and aquatic weeds. *Plant Science Today*. 2026;13(sp5):01–14. <https://doi.org/10.14719/pst.15292>

Abstract

Aquatic ecosystems are increasingly threatened by invasive aquatic weeds and harmful algal blooms (HABs), driven by eutrophication, climate change and anthropogenic disturbances. These proliferations degrade water quality, disrupt ecosystem functioning, reduce biodiversity and cause major economic losses in fisheries and water management. Conventional control strategies, including chemical, mechanical and biological methods, often provide limited effectiveness and may generate secondary environmental impacts. Ultraviolet–C (UV-C) radiation has emerged as a promising non-chemical alternative for sustainable aquatic ecosystem management. This review evaluates the mechanisms underlying UV-C mediated control of aquatic weeds and HABs, including DNA photodamage, reactive oxygen species generation, photosynthetic inhibition and membrane disruption. These processes collectively suppress cellular growth and viability. Recent advances in UV-C technologies, such as UV-light-emitting diode (UV-LED) systems, mobile treatment platforms and advanced oxidation processes [UV/hydrogen peroxide (UV/H₂O₂), UV/ozone (UV/O₃) and UV/peroxymonosulfate (UV/PMS)], have enhanced treatment efficiency for algal suppression, cyanotoxin degradation and pollutant removal. Treatment efficiency is strongly dose-dependent, with higher UV-C fluence generally resulting in greater microbial inactivation, algal suppression and cyanotoxin degradation. The review further examines field applications, engineering considerations and the integration of UV-C with smart monitoring tools including artificial intelligence, remote sensing and autonomous systems for precision aquatic management. However, treatment efficacy remains strongly influenced by turbidity, dissolved organic matter, water depth and species-specific tolerance, while challenges related to energy demand, operational cost and ecological impacts on non-target organisms persist. Future research should focus on optimising low-energy and solar-assisted UV systems, improving reactor design and integrating UV-C into ecosystem-based management frameworks. Overall, UV-C radiation offers a versatile and environmentally compatible strategy for sustainable control of aquatic weeds and HABs.

Keywords: advanced oxidation processes; artificial intelligence; cyanotoxin degradation; eutrophication; remote sensing; sustainable aquatic management; ultraviolet-light-emitting diode systems

Introduction

Aquatic ecosystems worldwide are increasingly affected by the rapid proliferation of aquatic weeds and harmful algal blooms (HABs), both of which have intensified in recent decades due to anthropogenic disturbances and changing climatic conditions. Invasive aquatic macrophytes such as *Eichhornia crassipes* (Mart.) Solms are now reported from more than 140 countries, where their aggressive growth alters ecosystem structure, reduces native biodiversity and interferes with water use for irrigation, fisheries and domestic purposes (1). The problem is particularly severe in tropical and subtropical regions, where favourable environmental conditions promote rapid spread. Uncontrolled expansion of aquatic weeds has been widely reported in irrigation reservoirs, lakes and aquaculture systems, where it disrupts water flow, increases flood risk and creates serious management challenges (2). The HABs

have also increased in frequency, duration and geographic distribution, largely driven by eutrophication caused by excessive nutrient loading, rising temperatures and altered hydrological regimes (3, 4). Cyanobacterial blooms alone are estimated to affect approximately 8.6 % of the global water surface, reflecting the magnitude of the issue (4). These blooms are commonly associated with nutrient-enriched water bodies receiving agricultural runoff, industrial effluents and urban wastewater, indicating a strong link between human activities and aquatic ecosystem degradation.

The ecological consequences of these phenomena are substantial: dense mats of aquatic weeds reduce light penetration and dissolved oxygen, leading to hypoxic conditions and disruption of aquatic food webs (5, 6). The HABs produce cyanotoxins such as microcystins, anatoxins and cylindrospermopsin that pose serious risks to aquatic organisms and human health (3, 4). Aquatic weeds

also contribute to sedimentation, nutrient cycling alterations and habitat modification, further degrading ecosystem stability and biodiversity (2). These toxins can accumulate in fish tissues and persist even after cooking, increasing the risk of exposure through the food chain (3). In addition to ecological impacts, economic losses are considerable, with HABs causing nearly \$8 billion annually in global aquaculture losses, while aquatic weeds reduce fish productivity, obstruct navigation and increase management costs, particularly in regions dependent on fisheries and agriculture (1, 3).

Despite extensive research, the management of aquatic weeds and HABs remains challenging. Conventional approaches such as mechanical removal, chemical treatments and biological control often provide only short-term or localised solutions. Mechanical methods are labour-intensive, costly and difficult to sustain at large scales (1), whereas chemical treatments may introduce secondary pollution and negatively affect non-target organisms (6). Biological control, although environmentally compatible, frequently shows variable effectiveness and requires continuous monitoring (1). Moreover, physical removal methods often lead to regrowth and dispersal of weed fragments, further complicating long-term management (2). In the case of HABs, most existing strategies focus on mitigating bloom impacts rather than addressing the underlying causes such as nutrient enrichment and climate variability, thereby limiting long-term effectiveness (3, 4). In this context, ultraviolet-C (UV-C) radiation has emerged as a potential alternative approach for controlling aquatic weeds. The UV-C irradiation acts by damaging cellular DNA, disrupting photosynthesis and inducing oxidative stress, resulting in reduced algal viability and suppression of bloom formation (3). Unlike chemical treatments, it does not introduce additional contaminants into water bodies, making it an environmentally safer option. Advances in UV-C reactor technologies, including flow-through and mobile systems, have improved its applicability in aquatic environments; however, its effectiveness is influenced by factors such as turbidity, dissolved organic matter and nutrient concentrations. Accordingly, a critical assessment of UV-C radiation, including its mechanisms, technological developments and environmental implications, is essential to determine its potential as a sustainable and effective approach for managing aquatic weeds and HABs. For this review, relevant literature was collected from scientific databases including Scopus, Web of Science, PubMed and Google Scholar using keywords related to UV-C radiation, HABs, aquatic weeds, cyanotoxin degradation and advanced oxidation processes. Peer-reviewed articles and recent studies focusing on UV-C mechanisms, treatment technologies, ecological impacts and aquatic management applications were primarily considered. Studies were selected based on scientific relevance and contribution to understanding UV-C applications in aquatic ecosystem management.

Ecology and drivers of aquatic weeds and harmful algal blooms

Aquatic weeds and HABs represent two major biological components of degraded aquatic ecosystems, each exhibiting distinct ecological characteristics but often driven by similar environmental conditions. Aquatic weeds are commonly classified by growth form into free-floating (e.g., *E. crassipes*), submerged (e.g., *Hydrilla* spp.), emergent (e.g., *Typha* spp.) and marginal species (7, 8). By contrast, HABs are dominated by planktonic cyanobacteria and other bloom-forming microalgae capable of rapid population expansion under nutrient-rich conditions. Among these, free-floating

macrophytes are considered the most aggressive due to their rapid vegetative reproduction, with species like water hyacinth capable of doubling biomass within 5–15 days under favourable conditions (7, 9). The widespread occurrence of invasive aquatic plants such as *E. crassipes*, *Salvinia molesta* D.Mitch. and *Pistia stratiotes* L. across multiple continents further highlights their ecological dominance and global invasion potential (10). Similarly, HABs are dominated by cyanobacteria and other phytoplankton groups that proliferate rapidly under nutrient-rich conditions, forming dense surface blooms. These organisms possess adaptive advantages such as buoyancy regulation, nitrogen fixation and tolerance to variable environmental conditions, enabling them to outcompete other aquatic flora (4). The ecological interactions between aquatic weeds and algal communities are often competitive and synergistic. Excessive macrophyte growth reduces light penetration and alters nutrient cycling, while algal blooms further modify oxygen dynamics and water chemistry, leading to unstable and degraded ecosystem states. Invasive aquatic plants also exhibit strong adaptability to diverse environmental conditions, enabling them to colonise a wide range of aquatic habitats (7).

The primary drivers of both aquatic weed proliferation and HAB formation are eutrophication and climate change, which together accelerate ecosystem imbalance. Increased nutrient inputs, particularly nitrogen and phosphorus from agricultural runoff, domestic sewage and industrial effluents, promote excessive primary productivity and facilitate the dominance of invasive macrophytes and bloom-forming algae (4, 8). In many aquatic systems, nutrient enrichment from urban wastewater and agricultural discharge has been directly linked to dense weed infestations and rapid spread of invasive species, often resulting in near-complete surface coverage of water bodies (11). Climate-related factors such as rising temperatures, altered precipitation patterns and prolonged stratification further enhance growth rates and bloom persistence, favouring species with high ecological plasticity (3). The major anthropogenic and climate-related drivers of eutrophication and their role in promoting aquatic weed proliferation and HABs are illustrated in Fig. 1.

At the ecosystem level, these processes result in severe consequences, including oxygen depletion, loss of biodiversity, disruption of trophic interactions and increased greenhouse gas emissions. Dense weed mats and algal blooms reduce water flow, impair hydropower and irrigation systems and create conditions conducive to disease vectors, thereby affecting ecosystem services and human wellbeing (8, 12). In addition, invasive wetland weeds such as *Eichhornia*, *Salvinia* and *Phragmites* have been associated with alterations in carbon cycling and greenhouse gas emissions, further contributing to climate-related impacts in aquatic ecosystems (13). The persistence of these drivers highlights the need for integrated ecological understanding to support effective and sustainable management strategies.

Constraints of conventional aquatic weed and algal control methods

Conventional approaches for managing aquatic weeds and HABs mainly rely on chemical, mechanical and biological control methods, yet each of these strategies is associated with significant limitations in terms of efficiency and ecological sustainability. Chemical control, particularly using herbicides such as diquat, remains one of the most widely adopted methods due to its rapid and visible effects on target vegetation (14). However, these

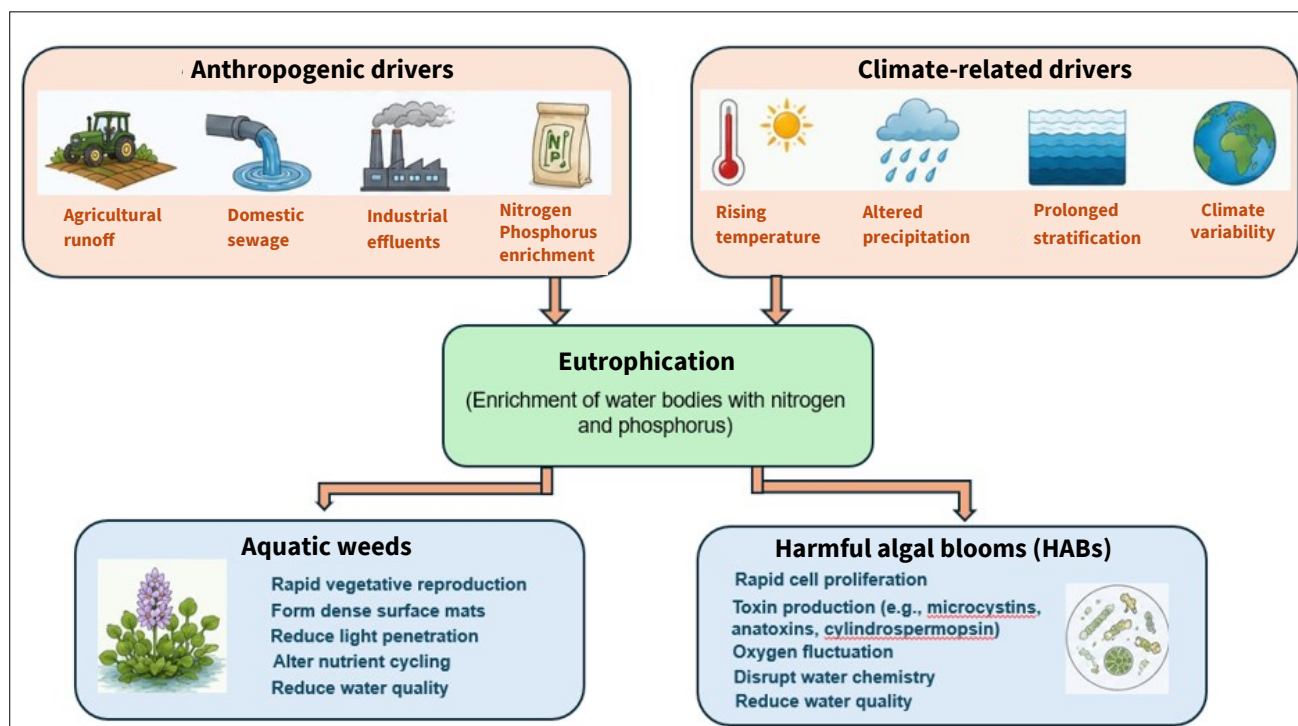


Fig. 1. Major anthropogenic and climate-related drivers of eutrophication and their role in promoting aquatic weed proliferation and harmful algal blooms in aquatic ecosystems.

herbicides are typically non-selective and can adversely affect non-target aquatic plants and organisms. Experimental studies have shown that even at recommended application rates (up to $1153 \mu\text{g L}^{-1}$), diquat can cause severe reductions in plant biomass and mortality in multiple aquatic plant species (15). It can also alter algal community composition and increase phytoplankton biomass following application. In addition, exposure to herbicide residues has been linked to oxidative stress responses in aquatic organisms, including disruption of antioxidant systems in model species such as zebrafish (16). Mechanical methods, including harvesting, dredging and cutting, offer immediate removal of biomass but are often costly and difficult to sustain, particularly in large or heavily infested water bodies. Moreover, fragmentation of plant material during mechanical removal can enhance vegetative propagation and facilitate downstream spread of invasive species, thereby reducing long-term effectiveness (17). Physical methods such as benthic barriers can reduce plant cover in localised areas. However, their performance is limited by sediment accumulation, gas build-up and maintenance requirements and they may also alter sediment biogeochemistry by promoting nutrient release (18).

Biological control methods, involving the use of herbivorous insects, fish, or pathogens, provide an environmentally compatible alternative but are often slow-acting, species-specific and dependent on ecological compatibility, which limits their widespread applicability. Although classical biological control has shown promise as a cost-effective and environmentally friendly strategy, it requires extensive host-specificity testing and long-term monitoring to avoid unintended impacts on non-target species (19). Furthermore, individual control approaches are often insufficient when used alone, as no single method can effectively manage large-scale infestations across diverse environmental conditions, highlighting the need for integrated strategies (20). Repeated herbicide use increases environmental contamination risk, promotes resistance development and often requires recurrent application because of rapid degradation, sediment binding, or

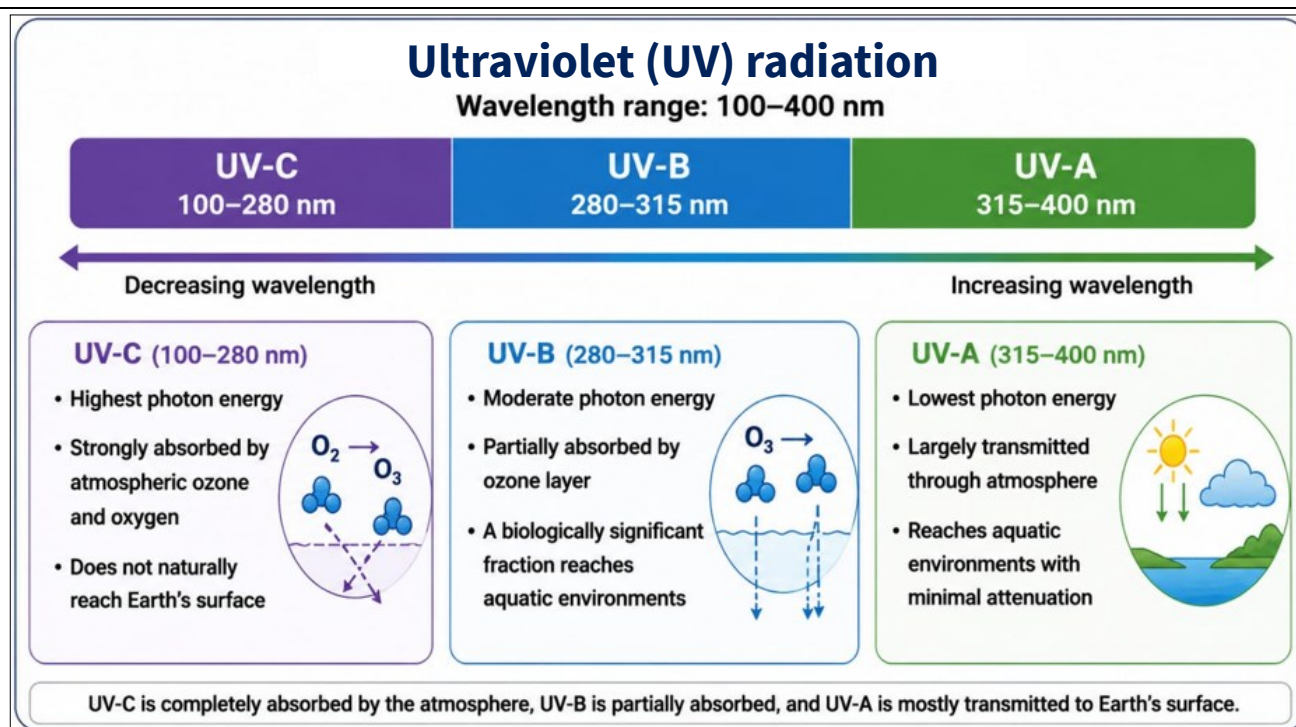
incomplete control (15). For instance, invasive species such as *E. crassipes* can rapidly regenerate and double their biomass within short periods, quickly offsetting the effects of control measures if not continuously managed (21). These combined limitations highlight the inadequacy of conventional approaches in addressing the complex and dynamic nature of aquatic weed and HAB proliferation, particularly under conditions of increasing eutrophication and climate change. Consequently, there is a clear need for alternative strategies that are both environmentally sustainable and capable of providing consistent, long-term control. A comparative evaluation of conventional and UV-C based control methods for aquatic weeds and HABs is presented in Table 1.

Fundamentals of ultraviolet radiation in aquatic environments

Ultraviolet radiation is a component of the electromagnetic spectrum with wavelengths ranging from 100–400 nm and is commonly classified into three regions: UV-A (315–400 nm), UV-B (280–315 nm) and UV-C (100–280 nm) based on their biological and physical properties (22). Among these, UV-C radiation possesses the highest photon energy and is strongly absorbed by atmospheric oxygen and ozone (O_3), preventing it from naturally reaching the Earth's surface (23, 24). In contrast, UV-B radiation is only partially absorbed by the stratospheric O_3 layer, allowing a biologically significant fraction to reach aquatic ecosystems, while UV-A is largely transmitted with minimal attenuation (23). The classification of UV radiation and its atmospheric transmission characteristics are illustrated in Fig. 2. Although UV radiation represents only a small fraction of total solar radiation, it exerts strong biological effects due to its interaction with cellular components such as DNA, proteins and lipids (23). Increased ultraviolet exposure associated with O_3 depletion has raised concerns regarding its ecological impacts on aquatic organisms, particularly algae and macrophytes, which form the foundation of aquatic food webs (25). The UV-C radiation has been shown to directly affect DNA integrity and disrupt physiological and biochemical processes in photosynthetic organisms, thereby influencing growth, productivity and survival (26).

Table 1. Comparative evaluation of conventional and ultraviolet-C based control methods for aquatic weeds and harmful algal blooms

Parameter	Chemical control	Mechanical control	Biological control	UV-C radiation	References
Mode of action	Herbicide-induced disruption of cellular metabolism	Physical removal through cutting and harvesting	Use of herbivores and pathogens	DNA damage, ROS generation and photosystem disruption	(20, 22)
Speed of response	Rapid action within days	Immediate but temporary control	Slow and gradual effect	Rapid cellular inactivation	(23, 22)
Environmental impact	High toxicity and residue persistence	Moderate disturbance to habitat	Generally eco-friendly	Minimal environmental impact	(16, 20)
Operational cost	Moderate to high due to repeated use	High labour and equipment cost	Moderate cost	Moderate energy-related cost	(24, 25)
Long-term effectiveness	Risk of resistance and repeated treatments	Frequent regrowth and reinfestation	Sustainable but slow establishment	Effective but influenced by water quality and exposure conditions	(12, 20)
Scalability	Suitable for large areas but environmentally constrained	Difficult in large infestations	Limited by ecological compatibility	Scalable through UV reactors and mobile treatment systems	(26, 27)
Key limitations	Resistance and pollution issues	Regrowth and scalability issues	Species-specific and slow action	Reduced efficiency in turbid water	(15, 22)


Fig. 2. Classification of ultraviolet (UV) radiation into UV-A, UV-B and UV-C based on wavelength range, atmospheric absorption and transmission to aquatic environments.

In aquatic environments, the penetration and effectiveness of UV radiation are strongly governed by the optical properties of water and associated constituents such as suspended particles and dissolved organic matter (DOM). Water itself acts as a natural filter, with shorter wavelengths (UV-B and UV-C) being rapidly attenuated compared to UV-A, resulting in steep vertical gradients of radiation with increasing depth (27). Factors such as turbidity, particulate matter and chromophoric dissolved organic matter significantly reduce UV transmission by scattering and absorption, thereby limiting its biological effectiveness. Experimental observations indicate that increased turbidity can substantially reduce UV-C disinfection efficiency, highlighting the importance of water clarity in determining treatment performance (28). In addition, physical environmental factors such as light intensity, temperature and hydrodynamic conditions play a critical role in regulating algal growth and UV interactions, as these factors influence photosynthesis, nutrient utilisation and bloom dynamics (29). The concept of UV dose or fluence, typically expressed as mJ cm^{-2} , is therefore essential for evaluating treatment efficiency, as it

integrates radiation intensity, exposure time and water quality conditions. Understanding these interactions between UV radiation and aquatic environments is crucial for optimising its application in the control of aquatic weeds and HABs.

Molecular and cellular mechanisms of ultraviolet-C induced damage

Ultraviolet-C radiation (200–280 nm) induces severe molecular damage primarily through direct interaction with nucleic acids and inhibition of cellular repair systems. DNA is one of the primary targets, where UV-C exposure leads to the formation of cyclobutane pyrimidine dimers and DNA strand breaks, disrupting replication and transcription processes. Experimental evidence shows that high UV-C doses ($150\text{--}300 \text{ kJ m}^{-2}$) can result in complete loss of DNA integrity and metabolic activity in algal cells such as *Chlorella minutissima* (30). Similarly, UV irradiation has been shown to significantly damage algal cell integrity, leading to membrane disruption and release of intracellular organic matter, indicating extensive cellular breakdown (31). In plant systems, UV exposure

induces DNA fragmentation, nuclear condensation and caspase-like protease activation, which are characteristic of programmed cell death (32). Due to its high photon energy, UV-C radiation overwhelms cellular repair mechanisms, leading to cell cycle arrest and apoptosis-like responses. A key secondary effect of UV-C radiation is the rapid generation of reactive oxygen species (ROS), including superoxide radicals, hydrogen peroxide and hydroxyl radicals, which amplify cellular damage. UV exposure has been reported to induce ROS production, protein degradation and oxidative damage in aquatic photosynthetic organisms, further impairing cellular metabolism (33). Elevated hydrogen peroxide (H_2O_2) concentrations and increased malondialdehyde (MDA) levels have been observed in UV-treated systems, reflecting lipid peroxidation and oxidative stress (30, 34).

The UV-C radiation also disrupts photosynthetic processes by targeting photosystem II (PSII) and the associated electron transport chain. Damage to the D1 protein of photosystem II (PSII) and impairment of electron transport reduce photochemical efficiency and chlorophyll fluorescence, leading to photoinhibition (35). Experimental observations indicate that UV exposure decreases pigment content, including chlorophyll a and b and reduces maximum quantum yield, thereby limiting photosynthetic energy conversion (34, 36). Studies on marine macroalgae have demonstrated that exposure to UV radiation significantly increases photoinhibition and impairs recovery of photosynthetic efficiency, particularly under prolonged high irradiance conditions (37). In species such as *Macrocystis pyrifera*, *Chondrus crispus* and *Ulva lactuca*, UV exposure reduces effective quantum yield and alters energy dissipation processes, reflecting damage to the photosynthetic apparatus and repair mechanisms. Disruption of electron transport also enhances ROS formation due to electron leakage, creating a feedback loop that intensifies oxidative stress.

The overall mechanism of UV-C-induced cellular damage follows a sequential pathway. Initially, UV-C photons are absorbed by

cellular DNA and other biomolecules, resulting in DNA lesions and disruption of replication and transcription processes. Simultaneously, excessive ROS generation causes oxidative damage to proteins, lipids and nucleic acids. Oxidative stress further impairs photosynthetic processes and enhances electron leakage from the photosynthetic apparatus, leading to additional ROS production. The resulting lipid peroxidation increases membrane permeability and destabilises cellular structures, while mitochondrial dysfunction disrupts energy metabolism and triggers programmed cell death pathways. Progressive deterioration of chloroplasts, mitochondria and cellular membranes ultimately results in loss of metabolic activity, disruption of cellular homeostasis and irreversible cell mortality. At the cellular level, UV-C radiation causes extensive membrane and organelle damage, leading to loss of structural integrity and cell viability. Lipid peroxidation results in increased membrane permeability and ion leakage, ultimately disrupting cellular homeostasis (30, 34). Mitochondrial dysfunction, including loss of membrane potential and impaired function, is a critical event associated with programmed cell death (32). Chloroplasts also undergo structural damage due to ROS accumulation, resulting in pigment degradation and reduced photosynthetic capacity. Collectively, DNA damage, oxidative stress, photosynthetic disruption and membrane degradation act synergistically to induce irreversible cellular injury and cell death, explaining the effectiveness of UV-C radiation in suppressing aquatic weeds and HABs.

The UV radiation also affects physiological performance and associated microbial communities of aquatic photosynthetic organisms. Studies on marine macroalgae (*Gelidium linguatum*) show that UV exposure reduces photosynthetic efficiency (F_v/F_m), indicating photoinhibition and impaired PSII recovery. It also alters epibiotic bacterial diversity and community composition, highlighting broader ecological impacts on host-microbe interactions and ecosystem stability (38). The molecular and cellular mechanisms of UV-C induced damage in aquatic organisms are illustrated in Fig. 3.

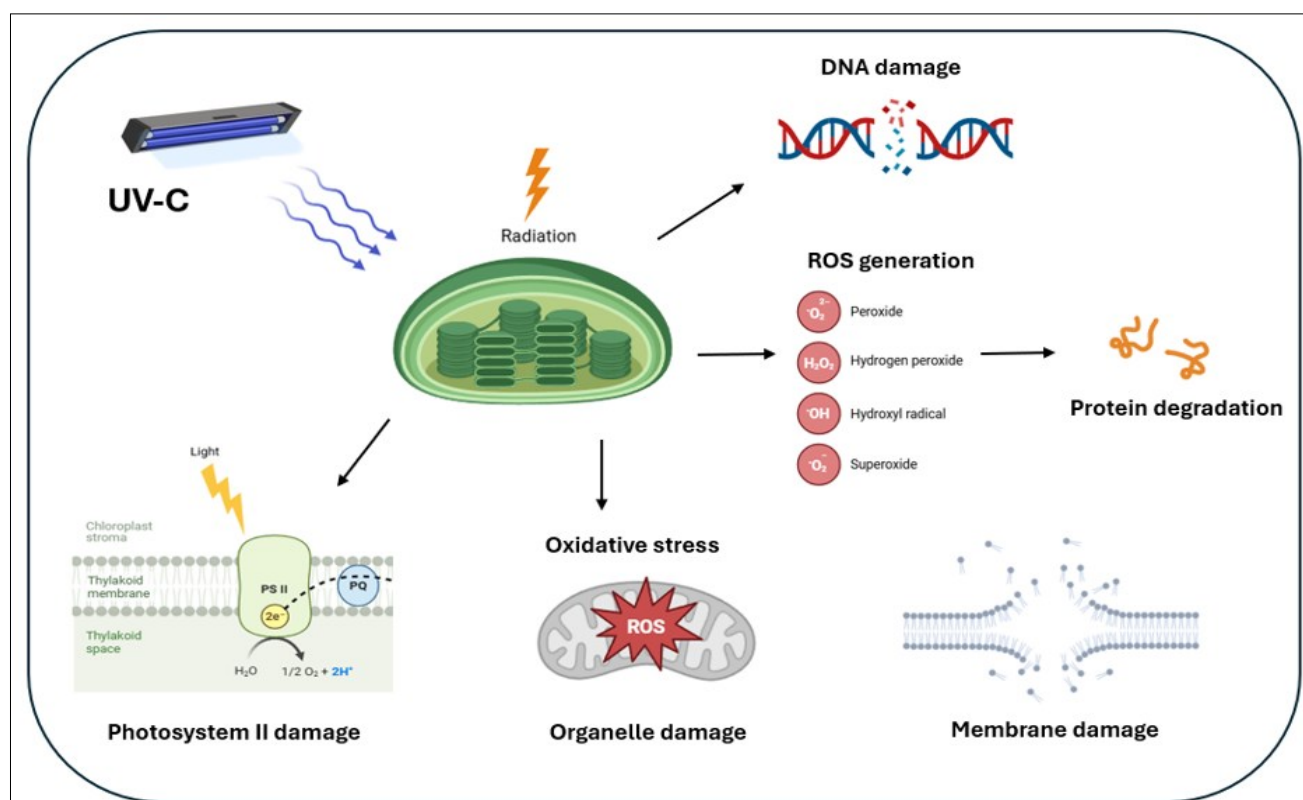


Fig. 3. Mechanisms of ultraviolet-C (UV-C)-induced damage in aquatic plants and algae.

Physiological and ecological responses to ultraviolet-C exposure

Exposure to UV-C radiation results in pronounced physiological stress in aquatic weeds and algal species, primarily expressed as growth inhibition and increased mortality. Experimental studies have shown that UV-C irradiation at 254 nm can reduce algal biomass by more than 80 % within short exposure periods, demonstrating its strong inhibitory effect (23). The effects of UV-C radiation on aquatic weeds and algal systems exhibit a dose-dependent response, where increasing UV-C dose leads to progressive growth inhibition, photosynthetic disruption and cellular damage, as shown in Fig. 4. Controlled experiments on *Chlorella vulgaris* show that UV-C exposure reduces algal concentrations by approximately 25–49 % within 1 hr, following an exponential decline pattern (39). Sub-lethal UV-C doses (20–200 mJ cm⁻²) can suppress cyanobacterial growth for several days, while higher doses (> 100 mJ cm⁻²) cause severe cell disintegration in more than 70 % of exposed cells, indicating a clear dose-dependent transition from growth inhibition to mortality (40). In addition, UV-C exposure has been shown to significantly reduce photosynthetic activity, with studies reporting a rapid decline of approximately 60 % in phytoplankton fluorescence immediately after exposure, reflecting impaired photosystem function (41). Declines in chlorophyll content ranging from 42 % to 77 % have also been observed under UV-C treatment, indicating disruption of pigment systems and reduced photosynthetic efficiency (42). However, under certain conditions, UV-C-induced oxidative stress can stimulate pigment production, including chlorophyll and carotenoids, as part of a protective metabolic response in microalgae such as *C. vulgaris* (43).

At the cellular and ecological levels, UV-C radiation induces structural alterations and species-specific responses that influence ecosystem dynamics. Morphological changes such as cell wall deformation, shrinkage and loss of cellular integrity have been observed following UV exposure, indicating direct physical damage

to algal cells (44). In both microalgae and macroalgae, UV-C treatment reduces biomass and alters growth patterns, with higher inhibition observed under increased exposure duration and shallow water conditions due to improved light penetration (45). Importantly, responses to UV-C are strongly dose-dependent and vary among species. Cyanobacteria such as *Microcystis aeruginosa* exhibit higher sensitivity, with rapid growth suppression even at lower doses, whereas green algae such as *Chlorella* and *Scenedesmus* tend to show comparatively greater tolerance (40). In higher plants and aquatic weeds, UV exposure can also modify biochemical pathways, particularly by influencing the synthesis of secondary metabolites involved in stress tolerance and defence mechanisms (46). Such biochemical adjustments may influence plant-herbivore interactions, competitive ability and broader ecological dynamics over time. Overall, the physiological and ecological responses to UV-C exposure demonstrate its effectiveness as a control strategy, while also emphasising the importance of species composition, environmental conditions and optimal dosage for achieving efficient and sustainable management outcomes. The effects of UV-C dose on aquatic weeds and algal systems are summarised in Table 2.

Adaptive and protective mechanisms under ultraviolet stress

Aquatic microorganisms, particularly cyanobacteria and algae, have evolved a range of adaptive mechanisms to withstand the damaging effects of UV radiation. One of the most important strategies is the synthesis of UV-screening compounds, including mycosporine-like amino acids (MAAs) and scytonemin, which absorb radiation in the UV range and prevent damage to cellular components. The MAAs typically absorb strongly between 310 and 360 nm and act as natural sunscreens, dissipating absorbed energy as heat without generating ROS (47). Similarly, scytonemin, an extracellular pigment found in cyanobacteria, forms a protective barrier that reduces UV penetration into cells, particularly under prolonged exposure conditions (48). In addition to these compounds, photosynthetic

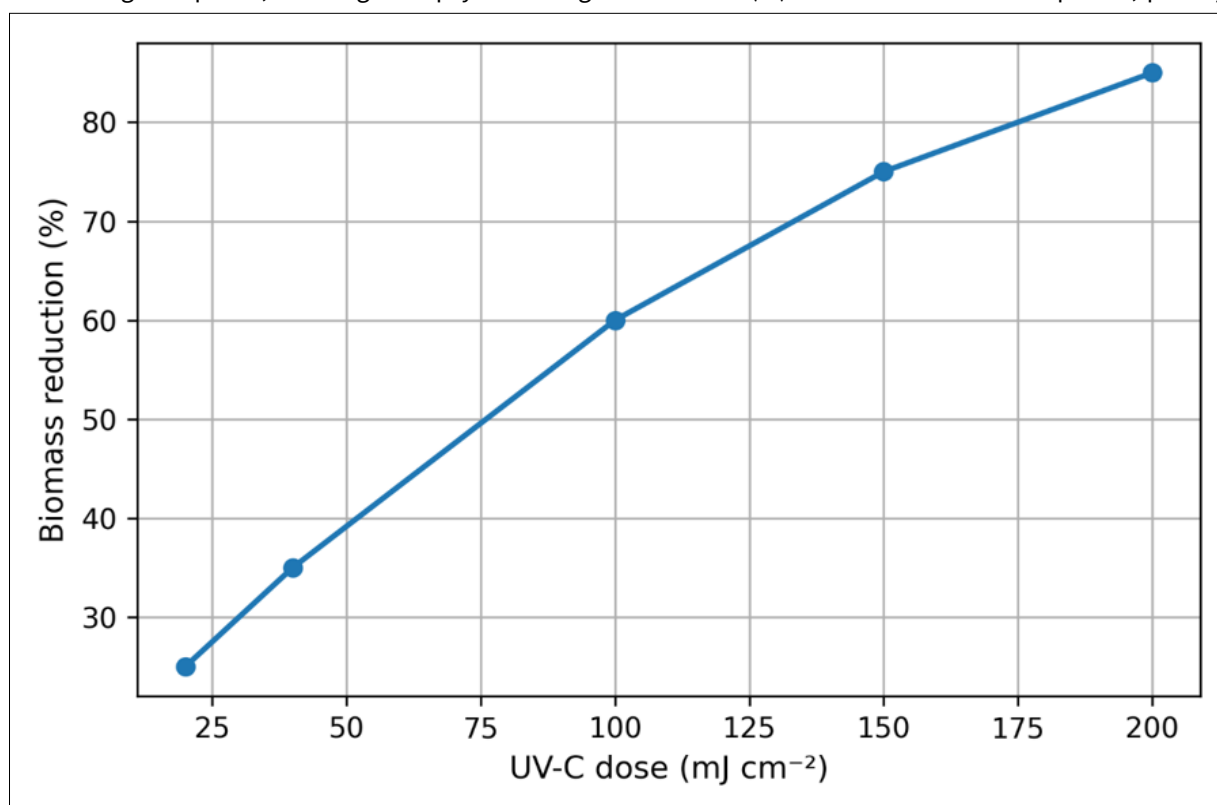


Fig. 4. Dose-response relationship between ultraviolet-C (UV-C) radiation and biomass reduction in aquatic weeds and algae.

Table 2. Effects of ultraviolet-C (UV-C) dose on aquatic weeds and algal systems

UV-C dose range	Biological effect	Observed response	Duration of impact	Reference
20–40 mJ cm ⁻²	Growth inhibition	Suppression of algal cell division and metabolic activity	Short-term (days)	(44)
100–200 mJ cm ⁻²	Photosynthesis disruption	40–77 % reduction in chlorophyll and pigment content	Medium-term (1–3 days)	(45)
> 100 mJ cm ⁻²	Cellular damage	More than 70 % cell disintegration and membrane damage	Long-term	(44)
150–300 kJ m ⁻²	Severe molecular damage	DNA fragmentation, metabolic shutdown, irreversible injury	Permanent	(36)
254 nm UV-C (short exposure)	Biomass reduction	Up to 80 % reduction in algal biomass	Rapid response (hours–days)	(23)
50–100 mJ cm ⁻²	Oxidative stress induction	Increased ROS production and lipid peroxidation	Short- to medium-term	(46)
200–300 mJ cm ⁻²	Growth suppression	Significant reduction in algal viability and photosynthetic activity	Long-term	(47)

organisms accumulate carotenoids and phenolic compounds that enhance photoprotection and antioxidant capacity. The synthesis of such protective metabolites is often induced under UV stress and varies among species and environmental conditions (26). Studies on aquatic bryophytes and other photosynthetic organisms further indicate that UV exposure can stimulate the production of UV-absorbing compounds and secondary metabolites, although the magnitude of response differs widely depending on species and habitat conditions (24). These biochemical adaptations are particularly important in shallow and clear water systems where UV exposure is high and continuous.

In parallel with photoprotective compounds, organisms activate antioxidant defence systems and DNA repair mechanisms to mitigate UV-induced damage. Enzymatic antioxidants such as superoxide dismutase, catalase and peroxidases play a crucial role in scavenging ROS generated during UV exposure, thereby reducing oxidative stress and cellular injury (26). Experimental evidence shows that external agents such as selenium nanoparticles can further enhance antioxidant defence by reducing hydrogen peroxide and lipid peroxidation levels, thereby alleviating UV-induced oxidative stress and improving cellular stability in microalgae such as *C. vulgaris* (49). DNA repair pathways, including photoreactivation mediated by photolyase enzymes and nucleotide excision repair, enable organisms to reverse UV-induced lesions such as pyrimidine dimers, particularly under light conditions (50). At the ecological level, biofilm formation provides an additional layer of protection, as extracellular polymeric substances create a physical barrier that reduces UV penetration and enhances community-level resistance. Mature biofilms exhibit higher tolerance to UV exposure compared to early-stage communities, indicating that structural complexity and microbial interactions contribute to resilience. These responses are often species-specific and strongly influenced by environmental conditions, highlighting the complexity of adaptive strategies in aquatic ecosystems. Overall, the integration of biochemical, physiological and ecological mechanisms enables aquatic organisms to survive under UV stress, although these protective systems may be insufficient under high-intensity UV-C exposure used for control applications.

Ultraviolet-C treatment technologies and process engineering

The UV-C treatment technologies have become a key component of modern water and wastewater disinfection systems due to their ability to inactivate microorganisms without the addition of chemicals. Conventional UV-C reactor systems generally use low-pressure mercury lamps emitting at 254 nm or

medium-pressure lamps with broader emission spectra. These systems are designed as batch or continuous-flow reactors, where water passes through quartz sleeves surrounding the UV source to ensure effective exposure. The performance of such systems is largely governed by UV fluence (dose), exposure time and reactor hydraulics. In most water treatment applications, UV doses in the range of 20–40 mJ cm⁻² are sufficient to achieve significant microbial inactivation, although higher doses are often required for resistant microorganisms and algal toxins (51). Experimental studies show that UV-C irradiation can achieve >6-log inactivation of bacteria such as *Escherichia coli* under controlled conditions (52). The efficiency of UV-C systems can be further enhanced through integration with advanced oxidation processes such as UV/hydrogen peroxide (UV/H₂O₂), where the generation of hydroxyl radicals improves the degradation of organic contaminants and cyanotoxins (53, 54). For instance, degradation of anatoxin-a exceeding 70 % has been reported at UV doses around 200 mJ cm⁻² in the presence of H₂O₂ (55). Despite their effectiveness, conventional systems face limitations including high energy consumption, mercury-related environmental concerns and operational challenges such as fouling of quartz sleeves, which can reduce UV transmission and overall treatment efficiency.

Recent technological advancements have led to the development of UV-LED systems, which offer improved flexibility and sustainability compared to traditional mercury-based lamps. The UV-LEDs emit radiation at specific wavelengths (typically 255–285 nm), allowing optimisation for target organisms and enhanced treatment performance. Studies indicate that UV-LED systems follow predictable dose–response kinetics, with D₁₀ values often below 10 mJ cm⁻² for sensitive microorganisms, enabling efficient reactor design and scaling (55). In addition, UV-LED systems operating at around 280 nm can achieve comparable or higher disinfection efficiency than conventional 254 nm lamps while reducing energy consumption by approximately 25–40 % (24). Effective reactor design requires careful consideration of parameters such as fluence rate, water turbidity, UV transmittance, flow conditions and reactor geometry, all of which influence light penetration and dose distribution (24). Reduction equivalent fluence, a metric describing biologically effective delivered dose, is commonly used to scale laboratory findings to full-scale systems and ensure uniform exposure. Field-scale innovations further demonstrate the applicability of UV-C technology, where controlled UV doses (10–50 mJ cm⁻²) can suppress or inactivate cyanobacterial blooms while minimising impacts on non-target organisms (26). Furthermore, hybrid treatment systems combining UV with

processes such as coagulation, ozonation and adsorption have been explored to improve removal of organic matter and contaminants, although UV-C remains primarily effective for microbial inactivation rather than bulk pollutant removal (56). Overall, advances in UV-LED technology, reactor engineering and field-scale deployment are enhancing the efficiency, adaptability and sustainability of UV-C-based treatment systems for aquatic environments.

Applications of ultraviolet-C in aquatic systems

The UV-C radiation has emerged as a promising tool for controlling HABs and aquatic microorganisms due to its strong inhibitory effects on cellular processes. Comprehensive reviews indicate that UV-C irradiation can effectively suppress microalgal growth through damage to nucleic acids, photosynthetic systems and cellular metabolism, with suppression strongly correlated to UV dose and environmental conditions (22). In practical applications, UV-C systems have been deployed in flow-through reactors and mobile platforms for *in situ* treatment of eutrophic water bodies. Field-scale evidence demonstrates that UV-C-based technologies can effectively reduce cyanobacterial abundance and associated toxin levels. For instance, UV-C treatment using a mobile boat system reduced microcystin concentrations by approximately 31–50 % and suppressed cyanobacterial growth for several days following treatment, highlighting its potential for rapid and chemical-free bloom management (26). The field-scale application of UV-C radiation using mobile or floating treatment platforms enables *in situ* control of aquatic weeds and HABs, as illustrated in Fig. 5. In addition, UV-C radiation is widely used for water disinfection, where relatively low doses (20–40 mJ cm⁻²) are sufficient to inactivate bacteria, viruses and protozoa, making it an essential component of drinking water and wastewater treatment systems. Beyond microbial control, UV-C radiation plays a significant role in the photodegradation of herbicides and organic pollutants in aquatic environments. Many herbicides, including diuron, fluroxypyr and clopyralid, undergo UV-induced photolysis, resulting in molecular

transformation and eventual mineralisation, with degradation rates strongly influenced by water chemistry and pH (57). For example, UV irradiation has reduced glyphosate concentrations by > 90 % while markedly lowering aquatic toxicity (58). Photodegradation also governs the environmental fate of aquatic herbicides such as diquat and fluridone, where UV exposure accelerates transformation via oxidation, demethylation and ring-cleavage pathways, thereby reducing persistence in the water column (59, 60). Field-based studies further demonstrate that UV-C-based interventions can contribute to substantial reductions in aquatic plant biomass, with treatment efficiencies approaching 75 % under controlled lagoon conditions (61). At the ecosystem level, integrated management approaches combining biological control, monitoring tools and emerging technologies such as UV-based treatments have been shown to enhance long-term effectiveness in aquatic weed management (62). Collectively, these applications highlight the dual role of UV-C radiation in both biological control and chemical pollutant degradation, supporting its potential as a sustainable and integrated strategy for aquatic ecosystem management.

Advanced oxidation and hybrid ultraviolet-based treatment systems

Advanced oxidation processes (AOPs) integrated with UV-C radiation have emerged as highly efficient strategies for the degradation of recalcitrant organic pollutants and cyanotoxins in aquatic systems. These processes operate through the *in situ* generation of highly reactive radicals, primarily hydroxyl radicals (•OH) and sulfate radicals (SO₄•⁻), which possess strong oxidation potentials and enable rapid, non-selective degradation of complex organic compounds. In UV-C/H₂O₂ systems, photolysis of hydrogen peroxide produces •OH that significantly enhance degradation efficiency compared to direct UV photolysis, with studies reporting up to 97 % degradation of anatoxin-a and a 4–5 fold improvement in reaction rates (63). Similarly, UV-C/PMS systems generate SO₄•⁻ with high redox potential (E° ≈ 1.82 V), achieving more than 99 % degradation

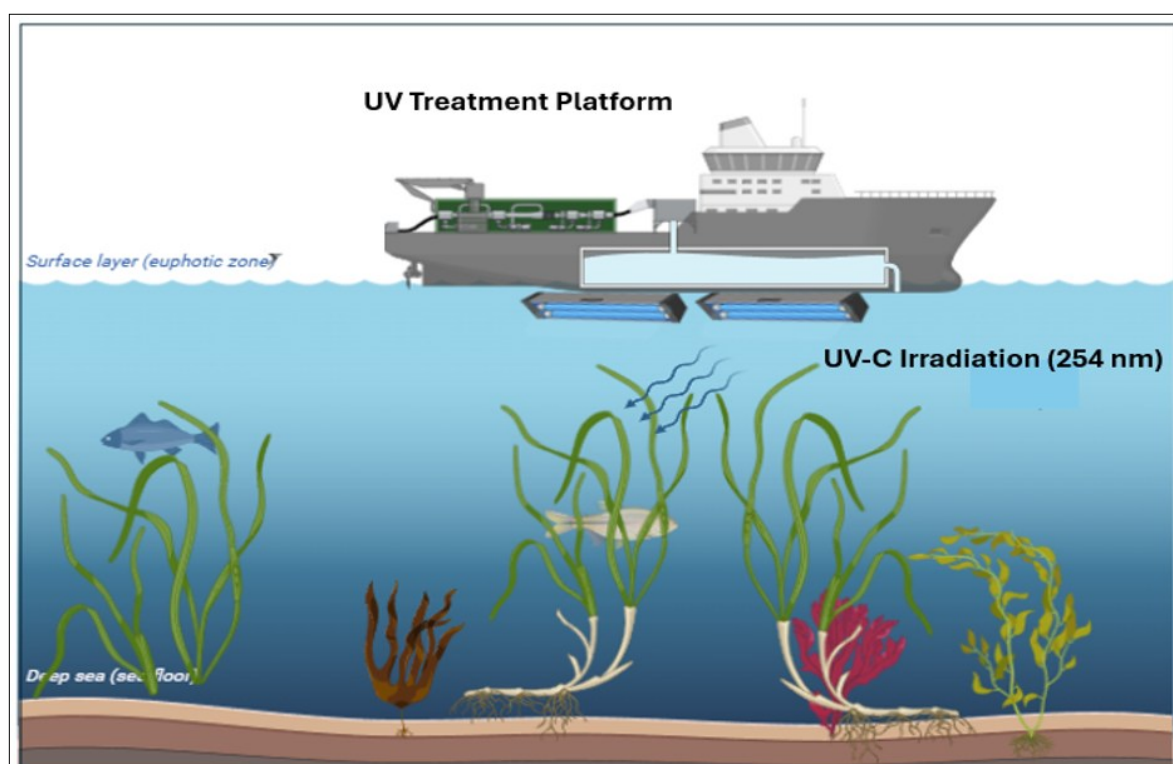


Fig. 5. Conceptual illustration of a mobile field-scale UV-C treatment system designed for the targeted suppression of aquatic weeds and harmful algal blooms in natural water bodies.

of cyanotoxins within short reaction times (64). In UV/O₃ systems, the combined action of O₃ decomposition and UV photolysis enhances radical production and improves mineralisation efficiency, with pollutant removal efficiencies increasing by approximately 40 % compared to ozonation alone while also reducing toxicity and antibiotic resistance risks (65).

Beyond pollutant degradation, UV-C radiation has also been explored for direct ecological applications such as aquatic weed control, offering a non-chemical alternative to conventional methods. A UV-C-based system operating at a wavelength of approximately 254 nm has been developed to control invasive aquatic vegetation by exposing submerged plants to high-energy radiation, leading to cellular damage and growth inhibition (66). This approach involves positioning UV-C light sources near aquatic plants, enabling effective irradiation in shallow water environments while avoiding issues associated with herbicides and mechanical removal, such as environmental contamination and fragmentation-induced spread.

In addition to standalone AOPs, hybrid UV-based treatment systems combining UV-C with coagulation, adsorption, photocatalysis and chemical oxidants have been developed to enhance treatment efficiency and operational feasibility. The UV-C-driven photocatalytic systems, particularly those employing titanium dioxide (TiO₂), significantly improve degradation kinetics of micropollutants, with reported rate constants reaching up to $1.53 \times 10^{-1} \text{ min}^{-1}$ under optimised conditions (67). Combined UV-C/sodium hypochlorite (UV-C/NaOCl) and UV-C/TiO₂ systems demonstrate superior performance compared to direct photolysis. Furthermore, UV-based AOPs contribute to biofilm control by generating ROS that disrupt microbial structures and inhibit regrowth (68). The efficiency of these systems is strongly influenced by operational parameters such as oxidant concentration, UV fluence, pH and water matrix composition, where carbonate and bicarbonate ions can act as radical scavengers and dissolved organic matter may either enhance or inhibit photodegradation depending on its characteristics (64). Overall, advanced oxidation and hybrid UV-based systems provide a versatile, efficient and sustainable

approach for the removal of aquatic pollutants and the management of invasive aquatic vegetation, highlighting their significant potential in modern water treatment and ecosystem management. The major UV-C based advanced oxidation systems and their treatment performance are summarised in Table 3.

Integrated and smart management approaches

The management of aquatic weeds and HABs increasingly relies on integrated strategies that combine physical, chemical, biological and UV-based approaches to improve efficiency and sustainability. The UV-C treatment can be effectively combined with reduced chemical inputs to minimise herbicide usage while maintaining control efficiency, thereby lowering risks to non-target organisms. Complementary biological approaches, including microbial degradation and herbivory, may provide longer-term suppression of invasive species (4). Integrated aquatic management frameworks emphasise coordinated steps including monitoring, targeted intervention and post-treatment evaluation, enabling site-specific treatments and improved resource efficiency (68, 69).

Recent advancements in remote sensing technologies have significantly enhanced large-scale monitoring. Unmanned aerial vehicles (UAVs) and satellite imagery enable accurate detection and classification of aquatic vegetation, often achieving classification accuracies greater than 90 % (70). In addition, advances in detection and monitoring techniques incorporating machine learning, deep learning and remote sensing have significantly improved the identification and tracking of invasive aquatic species such as *E. crassipes*, enabling more efficient and scalable ecosystem management (9). In parallel, artificial intelligence (AI) and machine learning models have transformed predictive aquatic ecosystem management, with hybrid frameworks integrating internet of things (IoT)-based environmental data achieving approximately 94 % accuracy in assessing plant health and forecasting ecological dynamics (71). Drone-assisted systems equipped with real-time object detection algorithms further enable precise identification and targeted treatment of invasive weeds, reducing labour requirements and enhancing operational efficiency (72).

Table 3. Advanced ultraviolet-C (UV-C)-based oxidation systems and their performance in aquatic treatment

System	Reactive species generated	Mechanism of action	Treatment efficiency	Reference
UV/H ₂ O ₂	Hydroxyl radicals (•OH)	Photolysis of hydrogen peroxide leading to non-selective oxidation of organic compounds	Up to 97 % degradation of cyanotoxins (anatoxin-a)	(65)
UV/O ₃	Hydroxyl radicals (•OH) and ozone intermediates	Ozone decomposition under UV radiation enhances radical formation and oxidation	~40 % higher removal compared to ozonation alone	(66)
UV/PMS	Sulfate radicals (SO ₄ • ⁻)	Activation of peroxymonosulfate producing strong oxidising radicals	> 99 % degradation within 10 min	(67)
UV/TiO ₂	Hydroxyl radicals (•OH)	Photocatalytic activation of TiO ₂ generating reactive species for pollutant breakdown	90 % pollutant degradation efficiency	(25)
UV-LED systems	UV photons (targeted wavelengths)	Direct photolysis with optimised wavelength emission	25–40 % energy saving with comparable efficiency	(24)
UV/Cl ₂	Hydroxyl radicals (•OH) and reactive chlorine species	UV activation of chlorine enhances oxidation and disinfection performance	Effective toxin degradation and microbial inactivation	(59)
Vacuum-UV (VUV)	Hydroxyl radicals (•OH) and hydrated electrons	Direct water photolysis generating highly reactive oxidising species	Efficient degradation of anatoxin-a and organic pollutants	(60)

UV/H₂O₂: ultraviolet/hydrogen peroxide; UV/O₃: ultraviolet/ozone; UV/PMS: ultraviolet/p peroxymonosulfate; UV/TiO₂: ultraviolet/titanium dioxide; UV-LED: ultraviolet light-emitting diode; UV/Cl₂: ultraviolet/chlorine; Vacuum-UV (VUV): vacuum ultraviolet; •OH: hydroxyl radical; SO₄•⁻: sulfate radical.

Autonomous robotic platforms and unmanned watercraft equipped with hydroacoustic sensing and machine learning have demonstrated high efficiency in aquatic weed identification and management, achieving classification accuracies of up to 99.06 % (73). Additionally, emerging UV-based technologies such as mobile treatment vessels enable real-time suppression of cyanobacterial blooms without chemical inputs, supporting rapid and targeted intervention strategies (26). The integration of these advanced technologies with IoT-based sensor networks for continuous monitoring of key water quality parameters such as pH, turbidity, temperature and dissolved oxygen allows real-time ecosystem assessment and early warning systems, facilitating proactive and adaptive management. Overall, these integrated and smart approaches provide a scalable, data-driven framework for sustainable aquatic ecosystem management, enhancing control efficiency while minimising environmental impacts and supporting long-term ecological resilience.

Limitations, risks and environmental considerations

Water quality constraints

The effectiveness of UV-C radiation in aquatic systems is strongly influenced by water quality parameters, particularly turbidity, dissolved organic matter and light attenuation. High turbidity and suspended particles reduce UV penetration, resulting in uneven dose distribution and decreased treatment efficiency. The major environmental and operational factors influencing the efficiency of UV-C radiation in aquatic systems are summarised in Table 4. In eutrophic and wastewater systems, reduced UV transmittance limits the performance of UV-based processes and often necessitates higher radiation doses to achieve desired outcomes (24). Dissolved organic matter can further interfere by absorbing UV radiation and scavenging reactive species, thereby reducing the efficiency of advanced oxidation processes.

Energy and economic limitations

Energy requirements also represent a major constraint, as conventional UV-C systems operate with relatively low energy efficiency and require continuous power input, increasing operational costs. Although UV-LED technologies provide advantages such as mercury-free operation and improved flexibility, their higher capital costs and technological limitations remain barriers for large-scale implementation. Economic assessments indicate that while UV-C processes can achieve relatively low treatment costs depending on the application, energy consumption and maintenance significantly influence overall feasibility (25).

Regrowth and ecological resilience

Another important limitation is the potential for regrowth and ecological resilience of target species following UV-C treatment. Many aquatic weeds and algal species exhibit rapid recovery due to high reproductive capacity, physiological adaptability and protective mechanisms such as biofilm formation and production of UV-screening compounds. For instance, invasive species like *E. crassipes* can quickly recolonise treated areas if management interventions are not sustained, reducing long-term control effectiveness (12).

Risks to non-target organisms

The UV-C treatment may pose risks to non-target organisms and ecosystem processes. Aquatic organisms are highly sensitive to environmental disturbances and exposure to pollutants or treatment-related stress can induce oxidative stress, physiological impairment and disruption of ecological balance even at low concentrations (74). Interactions between contaminants, such as herbicides and their degradation products, may further enhance toxicity and complicate ecosystem responses. Moreover, aquatic vegetation can influence contaminant dynamics by absorbing or transforming pollutants, thereby altering exposure levels for other organisms (75).

Environmental and ecosystem considerations

Physical control methods used alongside UV-based approaches, such as benthic barriers, may also modify sediment conditions, including reduced oxidation-reduction potential, potentially leading to the release of nutrients and metals into the water column (18). These limitations highlight the need for careful evaluation of environmental risks, system-specific conditions and long-term sustainability when applying UV-C technologies in aquatic ecosystems.

Future perspectives and research directions

Energy-efficient and sustainable ultraviolet-C technologies

Future advancements in UV-C-based aquatic management are expected to focus on improving energy efficiency and sustainability through the development of low-energy and solar-assisted UV systems. Emerging approaches such as solar-driven photocatalysis and hybrid UV systems powered by renewable energy sources offer promising alternatives to conventional energy-intensive UV-C technologies. In advanced oxidation processes, combined UV/O₃ systems have demonstrated improved pollutant removal efficiency while maintaining relatively low energy requirements, with electrical energy demand reported at less than 1 kWh m³ for effective

Table 4. Factors influencing the efficiency of ultraviolet-C (UV-C) radiation in aquatic environments

Factor	Influence on UV-C process	Effect on treatment efficiency	Reference
Turbidity	Scattering and absorption of UV radiation by suspended particles	Reduces UV penetration and lowers disinfection efficiency	(34)
Dissolved organic matter (DOM)	Absorbs UV radiation and scavenges reactive species	Decreases radical availability and treatment performance	(69)
Water depth	Attenuation of UV intensity with increasing depth	Limits effective exposure zone and reduces efficacy	(33)
UV dose (fluence)	Determines total energy delivered per unit area	Controls the level of microbial inactivation and pollutant degradation	(57)
Exposure time	Influences duration of UV contact with organisms	Longer exposure generally increases treatment effectiveness	(45)
Species type	Variability in sensitivity among organisms	Leads to species-specific responses and treatment outcomes	(46)
Temperature	Affects physiological activity and stress response of target organisms	Influences UV sensitivity and recovery potential	(35)
Water flow and mixing	Determines distribution of UV exposure within the treatment zone	Affects contact efficiency and treatment consistency	(22)

contaminant degradation (69). These systems also enhance mineralisation of dissolved organic matter and support self-purification processes in aquatic environments. However, challenges remain in optimising energy consumption, reducing operational costs and ensuring consistent performance under variable environmental conditions. Future research should therefore prioritise energy-efficient reactor design, optimisation of operational parameters and integration of renewable energy sources to improve the sustainability and scalability of UV-based treatment technologies.

Artificial intelligence (AI) and precision aquatic management

The integration of AI, remote sensing and automated systems is expected to transform aquatic ecosystem management by enabling precision-based, real-time monitoring and control. Hybrid machine learning frameworks combined with IoT-based sensor networks have demonstrated strong predictive capabilities, achieving approximately 94 % accuracy in aquatic plant health assessment and ecological forecasting (75). Similarly, AI-driven drone systems equipped with real-time object detection models enable accurate detection and targeted removal of invasive aquatic weeds, improving operational efficiency while reducing labour requirements (76).

Research gaps and future priorities

Despite these advancements, large-scale field deployment remains a significant challenge due to environmental variability, high initial investment costs and the need for reliable integration of multi-source data. Key research gaps include the development of cost-effective monitoring systems, improved data fusion from satellite, drone and *in situ* sensors and a deeper understanding of long-term ecological impacts of UV-based interventions. Addressing these priorities will be critical for advancing sustainable, ecosystem-based aquatic management strategies that effectively integrate technological innovation with environmental protection. Future success of UV-C aquatic management will depend on coupling energy-efficient engineering, precision monitoring and ecological safety within adaptive management frameworks.

Conclusion

Ultraviolet (UV)-C radiation represents a promising and environmentally sustainable approach for controlling aquatic weeds and harmful algal blooms (HABs) through mechanisms such as DNA damage, oxidative stress and photosynthetic disruption. Recent advances in UV reactors, UV-light-emitting diodes (UV-LEDs) and advanced oxidation processes have improved treatment efficiency and broadened field applications. However, treatment performance is influenced by water quality, energy requirements and potential ecological impacts on non-target organisms. Future research should focus on optimising low-energy UV-C systems and integrating UV-based treatments with ecosystem-based management strategies for long-term aquatic ecosystem sustainability.

Acknowledgements

The authors gratefully acknowledge the support and academic environment provided by Tamil Nadu Agricultural University, Coimbatore, India, which facilitated the preparation of this review article.

Authors' contributions

VC conceptualised the review topic, conducted the literature survey and drafted the manuscript. TK conceived the study, provided overall guidance and critically revised the manuscript. KR contributed to the design and organisation of the manuscript and assisted in the interpretation of the literature. SM contributed to content development, particularly on ecological and biological aspects and reviewed the manuscript. VS contributed to the technological and molecular aspects and assisted in refining the manuscript. UM supported data compilation and interpretation related to plant physiological aspects. ME assisted in literature collection and manuscript formatting. PRJ contributed to data organisation and manuscript editing. All authors have read and approved the final manuscript.

Compliance with ethical standards

Conflicts of interest: Authors do not have any conflicts of interest to declare.

Ethical issues: None

Declaration of generative AI and AI-assisted technologies in the writing process

The authors used Grammarly as an AI-assisted tool to support language editing, improve readability and enhance the organisation of the manuscript. The tool was used solely for linguistic refinement and structural clarity and did not contribute to data analysis, interpretation or the generation of scientific conclusions. All scientific content and responsibility for the final manuscript remain with the authors.

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Peer review: Publisher thanks Sectional Editor and the other anonymous reviewers for their contribution to the peer review of this work.

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